SANJIVANI RURAL EDUCATION SOCIETY'S

SANJIVANI K. B. P POLYTECHNIC, KOPARGOAN

SELF LEARNING ASSESSMENT REPORT

ON TOPIC

"UNIT III: ERROR DETECTION AND CORRECTION"

COURSE TITLE: DATA COMMUNICATION AND COMPUTER NETWORK

COURSE CODE: 314318

UNDER THE GUIDANCE OF:

Ms. K. V. SHARMA

SUBMITTED BY:

Mast. Adwait Shroff [233]

DEPARTMENT OF COMPUTER TECHNOLOGY
Sanjivani K. B. P. Polytechnic, Kopargaon – 423603

Sanjivani Rural Education Society's

Sanjivani K. B. P. Polytechnic

Department of Computer Technology

CERTIFICATE

This is to certify that the self-learning assessment [SLA] report on "UNIT III: ERROR DETECTION AND CORRECTION"

Submitted By:
Mast. Adwait Shroff [233]

Student of Department of Computer Technology, Sanjivani K.B.P. Polytechnic, in partial fulfilment of the requirement for the subject DATA COMMUNICATION AND COMPUTER NETWORK (Subject Code: 314318), has been completed under the guidance of MS. K. V. SHARMA.

Date: 24/03/26
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Subject Teacher
MS. K. V. SHARMA

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Unit III

Topic: Error Detection vs Error Correction

Student Name	Adwait Paresh Shroff
Roll Number	233
Class	Second Year – Computer Technology
Semester	Fourth Semester
Subject	Data Communication & Computer Networks
Academic Year	2025-26

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1. Topic Title & Unit Number

Topic Name	Error Detection vs Error Correction
Unit Number	Unit III – Data Link Layer
Subject	Data Communication & Computer Networks
Academic Year	2024-25
Roll No.	33
Student Name	Adwait Paresh Shroff

2. Learning Objectives

After studying this topic, I will be able to:

- Understand what errors are in data communication and why they occur during transmission.
- Clearly explain the difference between error detection and error correction.
- Identify the methods used for error detection (like parity check, checksum, CRC) and error correction (like Hamming code).
- Understand when to use detection and when to use correction in real networks.
- Answer 2-mark, 5-mark, and 10-mark questions based on this topic in exams.

3. Introduction / Definition

Whenever data is sent from one device to another through a network, there is always a chance that the data gets changed or corrupted because of noise, interference, or hardware problems. These unwanted changes in data are called errors.

Now, there are two things we can do when an error happens –

- Just find out that an error occurred – this is called Error Detection.
- Find the error AND also fix it automatically – this is called Error Correction.

Error Detection: It is the process of identifying whether any error has occurred in the transmitted data. It does not fix the error, just finds it.

Error Correction: It is the process of not only identifying the error but also correcting it at the receiver's end itself, without asking the sender to retransmit the data.

Both these techniques are part of the Data Link Layer which is the second layer of the OSI model. The main goal is to ensure that data is received correctly at the destination.

4. Basic Concept Explanation

To understand detection and correction, we first need to know what redundancy is. Redundancy means extra bits that are added to the original data before sending it. These extra bits are not actual data but they help in checking whether the data is correct or not at the receiver side.

4.1 Error Detection – Basic Idea

- Sender adds some extra bits (called redundancy bits or check bits) to the data.
- The receiver uses these bits to check if any error has occurred.
- If it finds a mismatch, it knows an error has happened and it requests for retransmission.
- It cannot fix the error on its own – it can only detect it.

4.2 Error Correction – Basic Idea

- Sender adds more extra bits than in detection (more redundancy).
- These extra bits are arranged in a special way so that they can tell exactly which bit went wrong.
- The receiver uses these bits to both identify and fix the error without asking for retransmission.
- This is also called Forward Error Correction (FEC).

Think of it like this – in an exam, if a teacher only marks right or wrong answers, that is detection. But if the teacher actually corrects your answer and gives the right one back, that is correction.

5. Detailed Explanation / Working

5.1 Error Detection Methods

A) Parity Bit Method

- A single extra bit (parity bit) is added to the data.
- In even parity: total number of 1s in the data + parity bit = even number.
- In odd parity: total number of 1s = odd number.
- At receiver side, parity is checked. If it doesn't match, error is detected.
- Limitation: can detect only single-bit errors, not burst errors.

B) Checksum Method

- The data is divided into blocks of equal size.
- All blocks are added together (using binary addition).
- The sum is complemented and sent as the checksum along with the data.
- At receiver side, all blocks including checksum are added. If result is all 1s, no error.
- Used in transport layer protocols like UDP and TCP.

C) Cyclic Redundancy Check (CRC)

- It is the most powerful and commonly used error detection technique.
- The sender treats the data as a binary number and divides it by a predetermined divisor (called generator polynomial).
- The remainder of this division is appended to the data and sent.
- Receiver performs the same division. If remainder is zero, data is correct.
- Can detect burst errors very effectively.

5.2 Error Correction Method

A) Hamming Code

- Most well-known error correction technique.
- Redundant bits are placed at positions which are powers of 2 (1, 2, 4, 8...).
- Each redundant bit is responsible for checking certain data bits.
- At receiver side, the bits are checked and the position of error is found from the pattern.
- Then the bit at that position is simply flipped to correct the error.
- This is a single-bit error correction method.

B) Retransmission (ARQ – Automatic Repeat reQuest)

- This is an indirect form of error correction.
- When error is detected, receiver sends a Negative Acknowledgement (NAK) to sender.
- Sender retransmits the data.
- Used in protocols like Stop and Wait ARQ, Go-Back-N ARQ, Selective Repeat ARQ.
- This is not a true FEC – it requires retransmission from sender.

6. Diagram / Illustration

Diagram 1: How Error Detection Works

```
SENDER SIDE:
  Original Data  +  Redundancy Bits  =  Frame Sent
  [1011001]      +     [101]         =  [1011001101]  ---->>
Channel
CHANNEL (Transmission Medium with Noise):
  [1011001101]  -->  Noise / Interference  -->  [1011011101]
(bit flipped!)
RECEIVER SIDE:
  Received Frame  [1011011101]  -->  Apply same algorithm  -->
Remainder != 0
  ERROR DETECTED!  -->  Send NAK  -->  Request Retransmission
```

Diagram 2: How Hamming Code (Error Correction) Works

```
Data bits:        D1  D2  D3  D4
Redundant bits:   R1  R2      R4
Bit positions:    1   2   3   4   5   6   7
                  R1  R2  D1  R4  D2  D3  D4
At receiver: Calculate parity checks C1, C2, C4
If C4 C2 C1 = 011 (binary) = 3 (decimal)  -->  Error at bit
position 3
CORRECTION:  Flip the bit at position 3  -->  Error corrected!
```

Diagram 3: Detection vs Correction – Flow Comparison

Error Detection Flow	Error Correction Flow
1. Sender adds check bits	1. Sender adds redundancy bits
2. Data is sent over channel	2. Data is sent over channel
3. Receiver checks for errors	3. Receiver finds error location
4. Error found: Send NAK	4. Receiver fixes error itself
5. Sender retransmits data	5. No retransmission needed

7. Example / Application

Example 1 – Parity Bit (Error Detection)

Suppose we want to send the data: 1011001

- Count the 1s: there are four 1s.
- For even parity, we need even number of 1s. Four is already even, so parity bit = 0.
- Frame sent: 10110010 (parity bit appended at end)
- If receiver gets 10110110 – now there are five 1s which is odd – MISMATCH – Error detected!

Example 2 – CRC (Error Detection)

Data = 1101011011, Generator polynomial = 10011

- Step 1: Append 4 zeros to data (since divisor has 5 bits, so degree = 4).
- Step 2: Divide 11010110110000 by 10011 using XOR division.
- Step 3: Remainder (say 1110) is appended to original data.
- Step 4: Receiver does same division. If remainder is 0000, data is error-free.

Real-Life Applications

- Error Detection is used in: Wi-Fi frames, Ethernet frames, TCP/UDP packets, CD/DVDs.

- Error Correction is used in: RAM memory chips (ECC RAM), deep space satellite communication, mobile phone networks (4G/5G), QR codes.
- QR codes can still be read even if part of them is damaged – because of error correction codes built into them.
- NASA uses error correction in space probes because retransmission from space takes too long.

8. Advantages

Advantages of Error Detection

- Simple and easy to implement.
- Requires fewer extra bits (less overhead on the data).
- Sufficient for reliable channels where errors are rare.
- Techniques like CRC are very powerful in detecting most types of errors.
- Widely supported in most existing protocols.

Advantages of Error Correction

- No need to retransmit data – receiver corrects error on its own.
- Reduces network congestion since retransmissions are not needed.
- Very useful in one-way communication (like TV broadcast, satellite) where sender cannot be asked to resend.
- Works well even in noisy channels.
- Saves time and bandwidth in long-distance or space communication.

9. Limitations / Disadvantages

Limitations of Error Detection

- It can only find that an error occurred, cannot fix it.
- Retransmission is required which wastes bandwidth and time.
- Not useful when retransmission is not possible (one-way channels).
- Some detection methods like parity bit cannot detect even number of errors.

Limitations of Error Correction

- More redundancy bits are needed which increases overhead on the data.
- Increases complexity of hardware and software.
- Generally corrects only a limited number of errors (like single-bit correction in Hamming code).
- More expensive to implement compared to detection only.
- Not practical for every type of application due to higher processing power needed.

10. Comparison Table – Error Detection vs Error Correction

Parameter	Error Detection	Error Correction
Definition	Only identifies that error has occurred.	Identifies and also fixes the error.
Action on Error	Requests retransmission from sender.	Corrects error at receiver itself.
Redundancy Bits	Fewer extra bits needed.	More extra bits needed.
Overhead	Less overhead on data.	More overhead on data.
Complexity	Simpler to implement.	More complex to implement.
Examples / Techniques	Parity Bit, Checksum, CRC	Hamming Code, Reed-Solomon Code
Retransmission	Required when error is found.	Not required – self-correcting.
Cost	Less costly.	More costly.
Use Case	Wired LANs, internet protocols.	RAM chips, satellite, QR codes.
Channel Type	Suitable for reliable channels.	Suitable for noisy / unreliable channels.
Full Form	—	Also called FEC (Forward Error Correction).

11. Important Terms / Keywords

Keyword	Meaning
Error	An unintended change in one or more bits of transmitted data.
Error Detection	Finding out that an error has occurred in received data.
Error Correction	Finding the error and automatically fixing it at receiver side.
Redundancy	Extra bits added to data for the purpose of error checking.
Parity Bit	A single extra bit added to make number of 1s even or odd.
CRC	Cyclic Redundancy Check – a powerful error detection method using polynomial division.
Hamming Code	An error correction technique that can identify and correct single-bit errors.
FEC	Forward Error Correction – correction done by receiver without retransmission.

Keyword	Meaning
ARQ	Automatic Repeat reQuest – receiver asks sender to resend data when error is detected.
Checksum	Sum of all data blocks used for error checking at receiver side.
Burst Error	Multiple consecutive bits getting corrupted during transmission.
Single Bit Error	Only one bit in the data unit is changed from 0 to 1 or 1 to 0.

12. Summary / Conclusion

Errors are a common problem in data communication and happen due to noise and interference in the transmission channel. To deal with this, two approaches are used – error detection and error correction.

Error detection only finds the presence of an error and then the data is retransmitted. It is simple, uses less redundancy, and is good enough for reliable channels. Methods like parity check, checksum, and CRC are used for this.

Error correction, on the other hand, not only detects but also automatically corrects the error at the receiver side without needing retransmission. This is useful in noisy channels and one-way communication. Hamming code is the best example of this.

In simple words – detection says 'something went wrong' and correction says 'something went wrong, and I have already fixed it'. Both are important for reliable data communication and the choice between them depends on the type of channel and application.

PRACTICE QUESTIONS Unit III – Error Detection and Correction

Section A – Short Answer Questions (2 Marks Each)

Q1. Define Error Detection. Give one example of an error detection technique.

Q2. What is meant by Error Correction? How is it different from Error Detection?

Q3. What is a Parity Bit? Differentiate between Even Parity and Odd Parity.

Q4. What is redundancy in data communication? Why is it added?

Q5. Define CRC. In which layer of the OSI model is it commonly used?

Q6. What is a Burst Error? Give one example from real life.

Section B – Descriptive Questions (5 Marks Each)

Q7. Explain the working of the Checksum error detection method with a suitable example.

Q8. Describe the Parity Bit method of error detection. What are its advantages and limitations?

Q9. What is Hamming Code? Explain with a diagram how it places redundant bits and detects/corrects a single-bit error.

Q10. Compare Error Detection and Error Correction on any five parameters using a table.

Q11. Explain ARQ (Automatic Repeat reQuest). How does it act as an indirect method of error correction?

Section C – Long Answer Questions (10 Marks Each)

Q12. Explain CRC (Cyclic Redundancy Check) in detail. How does the sender calculate the CRC remainder and how does the receiver verify it? Illustrate with an example using data = 1101011011 and generator = 10011.

Q13. Describe all three error detection techniques – Parity Bit, Checksum, and CRC. Compare their effectiveness, complexity, and real-world applications.

Q14. Write a detailed note on Forward Error Correction (FEC). Explain Hamming Code with a step-by-step example showing placement of redundant bits, calculation of parity, detection of error position, and correction of the error.

Q15. Discuss the real-life applications of error detection and error correction. Why is error correction preferred over detection in satellite communication and QR codes? Explain with proper justification.

13. References

- Data Communications and Networking – Behrouz A. Forouzan (4th Edition)
- Computer Networks – Andrew S. Tanenbaum
- Lecture Notes – Data Communication & Computer Networks (Class Notes)
- MSBTE Study Material for Error Detection and Correction
- GeeksForGeeks – Error Detection and Correction (https://www.geeksforgeeks.org/)

– End of Self Notes –